The quasi-Riemann hypothesis result (zero-free region Re(s) > 7/8 for all Dirichlet L-functions) is a major number-theoretic advance, but it doesn't change the *computational* hardness of factoring or discrete logs. Similarly, the Unique Games Conjecture proof (if verified) nails down hardness of approximation for problems like Max-Cut and Vertex Cover — important for complexity theory, but not a direct path to breaking any deployed cipher.

The more realistic threat vector, as Vitalik Buterin and others have flagged, isn't a single famous conjecture being solved. It's the **cumulative acceleration of algorithmic research**: AI finding *new attack algorithms* (like the number field sieve was for RSA) that shave off security margins on lattice problems. Each such discovery doesn't "solve" a famous conjecture, but it erodes the concrete security parameter estimates that NIST PQC standards rely on. Justin Drake's "bunker mode" warning is really about this: the *pace* of cryptanalytic discovery accelerating, not a single theorem dropping.

Bottom line

If the pace holds, the most plausible near-term threat to encryption is **not** a famous conjecture being solved, but rather:

1. **Incremental lattice-attack improvements** that reduce concrete security of Kyber/Dilithium below NIST thresholds — no famous conjecture involved, just better algorithms.
2. **A classical shortcut for ECDLP or factoring** — the kind of "380-year-old trick" that Scientific American recently highlighted with Fermat factorization against weak RSA implementations.
3. **P vs NP** remains the nuclear option, but it's a different order of difficulty from anything in the 722-paper release.

The famous conjectures (BSD, full RH, Hodge) are important, but solving them would be a *structural* victory, not an *algorithmic* one. The crypto threat is in the algorithmic layer, which is where the real extrapolation risk lives.

Elaborate

Ask a follow-up question

AI-generated answer. Please verify critical facts.